Geog 414 Spatial Data Management

Welcome to the official website for the *Spatial Data Management* course (GEOG-414) that I teach in Fall 2023 at the University of Tennessee, Knoxville.

Get the Book

- **US English Full-Color Print Edition (446 pages): Available on Amazon (link)**
- **US English PDF Edition (441 pages): Available on Leanpub (link)**

Cite the Book

If you use this book in your research or teaching, please consider citing it as follows:
Wu, Q. (2025). *Spatial Data Management with DuckDB: From SQL Basics to Advanced Geospatial Analytics*. Independently published. PDF edition ISBN 979-8993859705; Print edition ISBN 979-8274710572. https://duckdb.gishub.org

Table of Contents

- **Preface**
  - **Introduction**
  - **Who This Book Is For**
  - **What This Book Covers**
  - **Getting the Most Out of This Book**
  - **Conventions Used in This Book**
  - **Downloading the Code Examples**
  - **Video Tutorials and Supplementary Resources**
  - **Community and Feedback**
  - **Acknowledgments**
  - **About the Author**
  - **Licensing and Copyright**
- **DuckDB Foundations**
  - **Getting Started with DuckDB**
  - **Essential SQL for Spatial Analysis**
  - **DuckDB Python Integration**
- **Spatial Data Operations**
  - **Loading Spatial Data Formats**
  - **Exporting and Converting Spatial Data**
  - **Geometry Operations and Functions**
  - **Spatial Queries and Relationships**
  - **Advanced Spatial Joins**
  - **Interactive Data Visualization**
  - **Working with Vector Tiles and PMTiles**
- **Real-World Geospatial Analytics**
  - **Analyzing the US National Wetlands Inventory**
  - **Analyzing Global Building Footprints**
  - **Analyzing NYC Taxi Data**
  - **Developing Interactive Dashboards with Voilà and Solara**

How to Run Code Examples

The code examples are organized into folders, each corresponding to a chapter in the book. The code examples are written in Python and can be run using MyBinder, Google Colab, or Docker.

Using MyBinder

The code examples can be run using MyBinder.

Using Google Colab

The code examples can be run using Google Colab.

Video Tutorials

Complementing the written content, this book is supported by a comprehensive series of video tutorials that walk through key concepts and provide additional examples: https://tinyurl.com/duckdb-spatial-videos.

The videos are designed to complement, not replace, the written material. They're particularly helpful for:

- **Visual learners who benefit from seeing code being written and executed**
- **Understanding complex concepts through multiple explanations**
- **Learning about the development workflow and best practices**
- **Seeing how to approach problems and debug issues**

The playlist is organized to follow the book's structure. You can watch them in order as you progress through the book, or jump to specific topics as needed.

The videos were created in Fall 2023 when I was teaching the Spatial Data Management course at the University of Tennessee. Although the course has concluded, the videos remain relevant and can be used as a reference for the book. Additional videos will be added in the future.

About the Author

Dr. Qiusheng Wu is an Associate Professor in the Department of Geography & Sustainability at the University of Tennessee, Knoxville. He is also an Amazon Scholar. Dr. Wu's research focuses on advancing open-source geospatial analytics through cloud computing and GeoAI. He is the creator and maintainer of several widely used open-source Python packages, including Geemap, Leafmap, SAMGeo, and GeoAI, which integrate cloud-based geospatial platforms with AI-powered analysis and visualization. Dr. Wu's work bridges remote sensing, Earth observation, and artificial intelligence to make large-scale geospatial data more accessible, reproducible, and intelligent for researchers, educators, and practitioners worldwide. His open-source projects can be found on GitHub at opengeos.

Licensing and Copyright

This book embraces the principles of open science and open education. To support transparency, learning, and reuse, the code examples in this book are released under a Creative Commons Attribution 4.0 International (CC BY 4.0) license. This means you are free to copy, modify, and distribute the code, even for commercial purposes, as long as appropriate credit is given.

Please attribute code usage by citing the book or linking to the GitHub repository:

Wu, Q. (2025). *Spatial Data Management with DuckDB: From SQL Basics to Advanced Geospatial Analytics*. Independently published. PDF edition ISBN 979-8993859705; Print edition ISBN 979-8274710572. https://duckdb.gishub.org
While the code is freely available, the text, figures, and images in this book are copyrighted by the author and may not be reproduced, redistributed, or modified without explicit permission. This includes all written content, custom diagrams, and embedded visualizations unless otherwise noted.
If you wish to reuse or adapt any non-code material from the book (for example, for teaching, presentations, or publications), please contact the author to request permission. This dual licensing approach helps balance open access to learning materials with the protection of original creative work. Thank you for respecting these terms and supporting the open-source geospatial community.

Instructor
Dr. Qiusheng Wu is an Associate Professor in the Department of Geography & Sustainability at the University of Tennessee, Knoxville. In addition, he holds positions as an Amazon Visiting Academic and a Senior Research Fellow at the United Nations University. Specializing in geospatial data science and open-source software development, Dr. Wu is particularly focused on leveraging big geospatial data and cloud computing to study environmental changes, with an emphasis on surface water and wetland inundation dynamics. He is the creator of several open-source packages designed for advanced geospatial analysis and visualization, including geemap, leafmap, and segment-geospatial. For a closer look at his open-source contributions, please visit his GitHub repository: opengeos.

Course Schedule

Week	Date	Topic	Assignment
1	Aug 23, 25	Syllabus and Course Overview Intro to Spatial Data Models	
2	Aug 28, 30, Sept 1	Intro to Python and Jupyter Notebook	Lab 1
3	Sept 4, 6, 8	Getting Started with Earth Engine Sept 4 No Class (**Labor Day**)	Lab 2
4	Sept 11, 13, 15	Using Earth Engine Data	Lab 3
5	Sept 18, 20, 22	Using Local Geospatial data	Lab 4
6	Sept 25, 27, 29	Visualizing Geospatial Data	Lab 5
7	Oct 2, 4, 6	Analyzing Geospatial Data	
8	Oct 9, 11, 13	Exporting Geospatial Data Oct 9-10 No Class (**Fall Break**)	Midterm

9	Oct 16, 18, 20	Earth Engine Applications	Lab 6
10	Oct 23, 25, 27	Intro to Spatial Databases and DuckDB	Lab 7
11	Oct 30, Nov 1, 3	SQL Basics and DuckDB Python API	Lab 8
12	Nov 6, 8, 10	Data Import and Export	Lab 9
13	Nov 13, 15, 17	Geometric Operations and Spatial Relationships	Lab 10
14	Nov 20, 22, 24	Spatial Joins Nov 22-24 No Class (**Thanksgiving**)	
15	Nov 27, 39, Dec 1	PMTiles and Geodatabase	
16	Dec 4, 6	Final Project Presentations	Final project
17	Dec 12	Tuesday, 10:30 am - 12:45 pm	Final exam

Note: The instructor reserves the right to revise, alter or amend this syllabus as necessary. Students will be notified in writing/email of any such changes.

Software
Software Tools

- Miniconda or Anaconda
- Visual Studio Code
- Git
- QGIS

Cloud Computing Platforms

- Google Earth Engine
- Google Colab
- Amazon SageMaker Studio Lab
- Microsoft Planetary Computer

Python Basics

- 1. Getting Started
- 2. Variables and Data Types
- 3. Introducing List
- 4. Working with Lists
- 5. Conditional Statements
- 6. Dictionaries
- 7. Looping
- 8. Functions
- 9. Classes

- 10. Files and Exceptions

Earth Engine

- 11. Introducing GEE and Geemap
- 12. Creating Interactive Maps
- 13. Using Earth Engine Data
- 14. Using Local Geospatial Data
- 15. Visualizing Geospatial Data
- 16. Analyzing Geospatial Data
- 17. Exporting Earth Engine Data
- 18. Making Maps with Cartoee
- 19. Creating Timelapse Animations
- 20. Building Interactive Web Apps
- 21. Earth Engine Applications

DuckDB

- 22. Introduction
- 23. SQL Basics
- 24. Python API
- 25. Data Import
- 26. Data Export
- 27. Working with Geometries
- 28. Spatial Relationships
- 29. Data Visualization
- 33. Spatial Joins
- 34. Visualizing PMTiles
- 35. Data Analysis

PostGIS

- 36. Installation
- 37. SQL Basics
- 38. Introduction to PostGIS
- 39. Working with Geometries
- 40. Spatial Relationships
- 41. Projecting Data
- 42. Using GeoPandas

Labs

- Submission Instructions
- Lab 1
- Lab 2
- Lab 3
- Lab 4
- Lab 5
- Lab 6
- Lab 7

- Lab 8
- Lab 9
- Lab 10
- Final Project